Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Question 6

Part A

$Al_{(s)} \rightarrow Al^{3+} + 3e^-$

Part B

$3Zn^{2+}_{(aq)} + 2Al_{(s)} \rightarrow 3Zn_{(s)} + 2Al^{3+}_{(aq)}$

Part C

For every 2 mol of Al dissolved, 3 mol of Zn attach to the electrode.

2 mol Al. $\frac{26.98g}{1\ mol} = 53.96g$ 3 mol Zn $\frac{65.38g}{1} = 196.14$

For every 53.96 g of Al dissolved, 196.14 g of Zn appears, making Zn the electrode with the most change in mass.

Part D The maximum voltage comes from Be(s),

$1.85 + (-0.76) = 1.09\ E^0$

Continue to Question 7.